

International Islamic University Chittagong
Department of Electrical and Electronic Engineering
(Project/Thesis Proposal)

Application for the Approval of Project/Thesis Proposal

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4. Date of Assigning Supervisor	09-09-2025
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5. Date of Proposal Submission	
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6. Title	An Ultra-Low-Power 10-Transistor 1-Bit Magnitude Comparator Using Gate Diffusion Input Technique
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7. Introduction Modern digital computing systems and portable electronic devices demand arithmetic blocks that consume minimal power while maintaining high execution speeds and compact silicon footprints [1]. Among these foundational blocks, the digital magnitude comparator is extensively utilized in microprocessors, sorting units, and control logic to evaluate the relative magnitude of two binary inputs (A and B), yielding three mutually exclusive conditions: $A > B$, $A = B$, and $A < B$ [2].

Standard complementary metal-oxide-semiconductor (CMOS) implementations of 1-bit comparators conventionally rely on standard logic gates (AND, OR, NOT, and XNOR), which often result in larger transistor counts, higher capacitive switching loads, and increased power dissipation [3]. To overcome these architectural bottlenecks, alternative circuit design methodologies such as Gate Diffusion Input (GDI) have emerged as powerful solutions for low-power applications [5]. The GDI technique enables complex Boolean functions to be realized using a remarkably reduced number of transistors by applying input signals directly to the gate, source, and drain terminals of MOS devices [6].

By leveraging the GDI technique, a standard 1-bit magnitude comparator can be optimized into an ultra-low-power 10-transistor configuration [7]. This layout significantly curtails static leakage and dynamic switching power compared to conventional CMOS equivalents [4]. This project focuses on designing, simulating, and evaluating this 10-transistor GDI comparator to comprehensively analyze its performance trade-offs in terms of power dissipation, propagation delay, and Power Delay Product (PDP) [8].

8. Literature Review Extensive research has been conducted on optimizing 1-bit magnitude comparators to enhance energy efficiency and reduce hardware complexity. Kumar and Kumar demonstrated an improved CMOS 1-bit comparator using a stacking technique in 180-nm technology, establishing that stack structures effectively suppress subthreshold leakage currents and improve the Power Delay Product (PDP) [1].

Similarly, Tailor *et al.* investigated multi-threshold CMOS (MTCMOS) and forced stack schemes to minimize power loss in 180-nm technology nodes [2]. Their findings emphasize the critical role of transistor sizing and threshold voltage management in balancing speed and power. Advancing further into novel device paradigms, Rajora *et al.* explored carbon nanotube field-effect transistor (CNTFET) architectures, utilizing complementary techniques to achieve high-speed comparison operations at ultra-low energy levels [3].

In parallel, alternative logic styles like Double-Pass-Transistor Logic and Gate Diffusion Input (GDI) have gained prominence for arithmetic circuit optimization. Morgado *et al.* originally formulated the GDI paradigm to facilitate low-power digital designs with reduced transistor counts [5], while Navi *et al.* successfully demonstrated its efficacy in arithmetic blocks like adders and multiplexers [6]. Golshan *et al.* extended GDI principles to high-performance magnitude comparators, proving substantial area and power reductions over conventional designs [7]. Furthermore, Saha *et al.* implemented rigorous transient and post-layout verification flows for advanced comparator structures [4], while Weste and Harris provided the foundational design rules for robust CMOS and alternative logic realizations [8].

The reviewed literature collectively validates the feasibility and advantage of utilizing GDI-based topologies for achieving ultra-low-power 1-bit magnitude comparators.

9. Problem Statement/Research Gap Traditional CMOS 1-bit magnitude comparators typically demand high transistor counts—frequently requiring 20 to 30 or more transistors depending on the logic gate configuration [1]. This high component count directly leads to increased input capacitance, higher dynamic switching power, and a larger silicon footprint [2].

While techniques such as transistor stacking and MTCMOS offer partial improvements, they often introduce fabrication complexity or fail to achieve optimal area reduction [7]. The primary research gap lies in designing an ultra-low-power 1-bit magnitude comparator using the Gate Diffusion Input (GDI) technique restricted strictly to a 10-transistor budget, ensuring minimal area and power consumption without severe signal degradation or excessive propagation delay [6, 8]. This study bridges this gap through rigorous SPICE-based simulation and comparative performance evaluation.

10. Objectives with Specific Aims General Objective

To design, simulate, and evaluate an ultra-low-power 10-transistor 1-bit magnitude comparator using the Gate Diffusion Input (GDI) technique [7].

Specific Objectives

1. To derive and verify the optimized Boolean expressions and truth table for a 1-bit magnitude comparator suitable for GDI cell integration [8].
2. To design the transistor-level schematic of the 10-transistor GDI comparator topology [7].
3. To perform transient SPICE simulations for functional verification across all input combinations under standard operating conditions [4].
4. To measure and evaluate performance metrics including average power consumption, propagation delay, Power Delay Product (PDP), and total transistor count [1, 2].

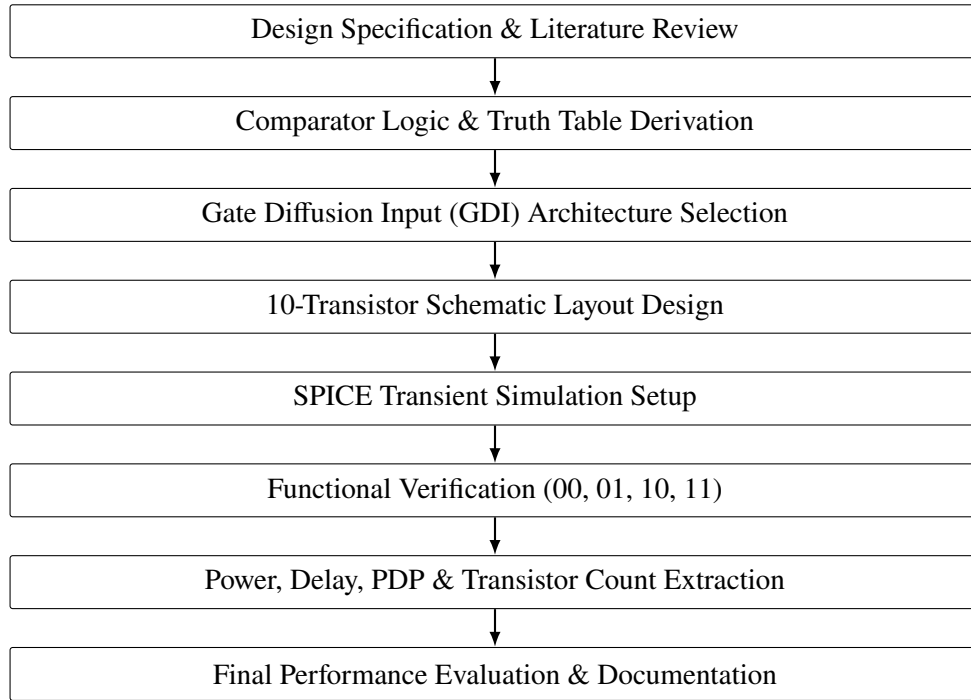
11. Proposed Methodology The methodology follows a structured VLSI design flow, moving from theoretical formulation to transistor-level simulation and metric extraction [8].

1. **Specification and Review:** Establish baseline supply voltages, capacitive load conditions, and performance evaluation standards based on established literature [1].
2. **Logic Derivation:** Formulate the optimized Boolean equations for the comparator outputs [8]:

$$G = A\bar{B}, \quad E = AB + \bar{A}\bar{B}, \quad L = \bar{A}B.$$

3. **Circuit Schematic Design:** Construct the 10-transistor architecture utilizing GDI logic cells to minimize active device count [7].
4. **Functional Verification:** Test the design via transient SPICE simulation across all input permutations (00, 01, 10, 11) to ensure correct logical functionality [4].
5. **Performance Extraction:** Measure average power dissipation and propagation delay using standard 50% transition thresholds, and compute the Power Delay Product ($PDP = P_{avg} \times t_p$) [2].

Methodology Framework



Functional Truth Table

A	B	$A > B$	$A = B$	$A < B$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

12. Gantt Chart The project follows a 12-month timeline, with the first four months leading up to the proposal approval and the remaining months dedicated to design, simulation, and thesis documentation.

Activities	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Topic selection and problem identification	✓	✓										
Literature review	✓	✓	✓	✓								
Proposal preparation and presentation		✓	✓	✓								
Requirement analysis and methodology development			✓	✓	✓							
1-bit comparator Boolean formulation				✓	✓							
10-T GDI circuit design					✓	✓	✓					
SPICE simulation and functional verification						✓	✓	✓				
Power and delay measurement							✓	✓	✓			
PDP and transistor-count analysis								✓	✓	✓		
Performance evaluation									✓	✓	✓	
Thesis writing and documentation									✓	✓	✓	✓
Thesis revision											✓	✓
Final submission, presentation and defense												✓

13. References

References

- [1] A. Kumar and M. Kumar, "Improved Design of CMOS 1-Bit Comparator With Stacking Technique," in *2017 2nd International Conference on Telecommunication and Networks (TEL-NET)*, 2017.
- [2] D. Tailor, T. Sharma, and M. Kumar, "Design of CMOS Based 1-bit Comparator with MTCMOS and Forced Stack Technique in 180 nm Technology," in *2019 IEEE*, 2019.
- [3] R. Rajora, K. Sharma, L. Gupta, A. Sachdeva, and A. Sharma, "Low-Power High-Speed CNTFET-based 1-bit Comparator Design using CCT and STT Techniques," in *Proceedings of the Second International Conference on Electronics and Renewable Systems (ICEARS)*, 2023, pp. 49–54, doi: 10.1109/ICEARS56392.2023.10084990.
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- [5] A. Morgado, R. Castro, et al., "Gate Diffusion Input (GDI): A new technique for low-power digital circuit design," in *IEEE/CAS Region 8 Conference (COLAM)*, 2002.
- [6] K. Navi, M. Sayedsalehi, R. Farazkish, and A. Hosseini, "A novel low-power full adder cell based on GDI technique," *Microelectronics Journal*, vol. 40, no. 10, pp. 1443–1448, 2009.
- [7] M. Golshan, K. Navi, and O. Hashemipour, "Low-power and high-speed magnitude comparator using GDI logic," *Journal of Circuits, Systems and Computers*, vol. 27, no. 5, p. 1850075, 2018.
- [8] N. H. E. Weste and D. Harris, *CMOS VLSI Design: A Circuits and Systems Perspective*, 4th ed. Boston: Pearson, 2011.

14. Justification of the Problem as Complex Engineering Problem This project qualifies as a complex engineering problem because it requires managing competing trade-offs in transistor-level circuit design, such as balancing power reduction against propagation delay and signal integrity within a restricted 10-transistor GDI framework.

Relevant Washington Accord Graduate Attributes (WA):

- **WA1 – Engineering Knowledge:** Application of semiconductor physics, digital electronics, and GDI logic principles.
- **WA2 – Problem Analysis:** Evaluation of power consumption versus speed trade-offs.
- **WA3 – Design and Development:** Developing an optimized 10-transistor comparator topology.
- **WA4 – Investigation:** SPICE simulation, verification, and measurement analysis.
- **WA5 – Modern Tool Usage:** Utilizing industrial SPICE simulators and EDA tools.
- **WA12 – Life-long Learning:** Reviewing contemporary low-power VLSI literature.

15. Budget (if any)

Item	Cost (BDT)
SPICE/EDA software	0
Internet and online resources	1500
Printing and binding	2000
Miscellaneous	1500
Total	5000

Signature of the Students

Signature of the Supervisor

Signature of the Convener, Thesis/Project Committee